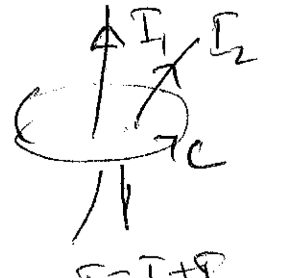
* Ampere's Theorem for magnetostatics (steady currents)

Let C be a closed loop.
Current $I_{enc.}$ flows through the surface within the loop. Then

$$\boxed{\oint_C \vec{B}\cdot\vec{dl} = \mu_0 I_{enc}}$$



$I_{encl} = I_1 + I_2$

This is the integral form of Ampere's Theorem.
Can use it to derive a differential form:

$$\oint_C \vec{B}\cdot\vec{dl} = \int_\Gamma (\vec{\nabla}\times\vec{B})\cdot d\vec{a} \quad \left(\begin{array}{c}\text{Stokes'}\\ \text{theorem}\end{array}\right)$$

Also $\mu_0 I_{enc.} = \mu_0 \int_\Gamma \vec{J}\cdot d\vec{a} = \int_\Gamma (\mu_0\vec{J})\cdot d\vec{a}$

Thus $\int_\Gamma (\vec{\nabla}\times\vec{B})\cdot d\vec{S} = \int_\Gamma (\mu_0\vec{J})\cdot d\vec{a}$

for ANY simply connected surface Γ

$\Rightarrow \quad \vec{\nabla}\times\vec{B} = \mu_0\vec{J}$

* Remember similar derivation of $\vec{\nabla}\cdot\vec{E} = 0$ from Gauss's flux theorem, using Gauss's divergence theorem.

* 'Magnetostatics in differential form	Electrostatics in differential form
$\vec{\nabla}\cdot\vec{B} = 0$	$\vec{\nabla}\cdot\vec{E} = \rho/\epsilon_0$
$\vec{\nabla}\times\vec{B} = \mu_0\vec{J}$ [will be corrected]	$\vec{\nabla}\times\vec{E} = 0$ [will be corrected]

corrected forms will be Maxwell-III & Maxwell-IV

REVIEW

	Charges & steady currents	Field acts on charge	How fields are created	Integral theorems ~~Relations~~
Electrostatics	$Q = \int \rho\, dV = \int \rho\, d^3r$ $Q = \int \sigma\, dS = \int \sigma\, da$ $Q = \int \lambda\, dl$	$\vec{F} = q\vec{E}$	$\vec{E} = \frac{q}{4\pi\epsilon_0} \frac{\widehat{(\vec{r}-\vec{r}')}}{\lvert\vec{r}-\vec{r}'\rvert^2}$ NO NAME $\simeq$ Coulomb's law + superposition princip.	GAUSS's theorem (Dielectric flux thm) $\oint \vec{E}\cdot d\vec{S} = \frac{1}{\epsilon_0} Q_{enc}$
Magnetostatics	$I = \int \vec{J}\cdot d\vec{S}$	$\vec{F} = q\vec{v}\times\vec{B}$ $d\vec{F} = I\, d\vec{l}\times\vec{B}$	$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l}\times\widehat{(\vec{r}-\vec{r}')}}{\lvert\vec{r}-\vec{r}'\rvert^2}$ Biot-Savart + superposition princip.	Ampere's theorem $\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$

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Electrostatics	$\vec{\nabla}\cdot\vec{E} = \rho/\epsilon_0$ (GAUSS)	$\vec{\nabla}\times\vec{E} = 0$	$\vec{E} = -\vec{\nabla} V$ (possible because $\vec{\nabla}\times\vec{E} = 0$ $\Rightarrow$ will need correction)	$V = -\int_{\vec{r}_0}^{\vec{r}} \vec{E}\cdot d\vec{r}$ Depends on $\vec{r}_0$!
Magnetostatics	$\vec{\nabla}\cdot\vec{B} = 0$	$\vec{\nabla}\times\vec{B} = \mu_0\vec{J}$ (AMPERE) $\Downarrow$ will need time-dependent correction	$\vec{B} = \vec{\nabla}\times\vec{A}$ possible because $\vec{\nabla}\cdot\vec{B} = 0$	$\vec{A}$ depends on "gauge" choice

V = electric potential / scalar potential

$\vec{A}$ = vector potential

* We also learned: $\vec{E} = \sigma \vec{J}$, $\vec{J} = \rho \vec{E}$, $V = RI$ where σ, ρ, R are material- & temperature-dependent.

* We learned the CONTINUITY EQUATION

$$\vec{\nabla}\cdot\vec{J} + \frac{\partial \rho}{\partial t} = 0$$

For steady current: $\frac{\partial \rho}{\partial t} = 0$, $\vec{\nabla}\cdot\vec{J} = 0$

* Is Ampere's law $\left(\vec{\nabla}\times\vec{B} = \mu_0 \vec{J}\right)$ consistent with the continuity equation?

Since $\vec{J} = \frac{1}{\mu_0}\vec{\nabla}\times\vec{B}$,

$$\vec{\nabla}\cdot\vec{J} = \frac{1}{\mu_0}\vec{\nabla}\cdot\left(\vec{\nabla}\times\vec{B}\right) = 0$$

$\Rightarrow$ consistent, only for steady currents, $\rho = $ const.

When $\frac{\partial \rho}{\partial t} \neq 0$, ~~[illegible]~~

not consistent! $\longrightarrow$ $\vec{\nabla}\times\vec{B} = \mu_0 \vec{J}$ has to be corrected

For consistency, one needs $\vec{\nabla}\times\vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

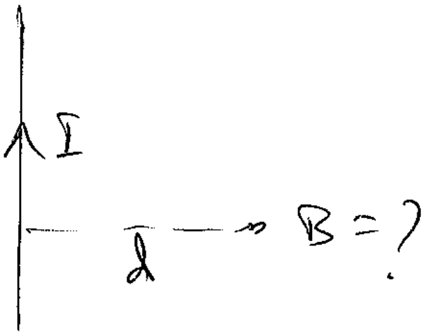
Because then $\vec{J} = \frac{1}{\mu_0}\vec{\nabla}\times\vec{B} - \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

so that $\vec{\nabla}\cdot\vec{J} = 0 - \epsilon_0 \frac{\partial}{\partial t}(\vec{\nabla}\cdot\vec{E}) = -\epsilon_0 \frac{\partial}{\partial t}\frac{\rho}{\epsilon_0} = -\frac{\partial \rho}{\partial t}$

Using Ampere's law/theorem

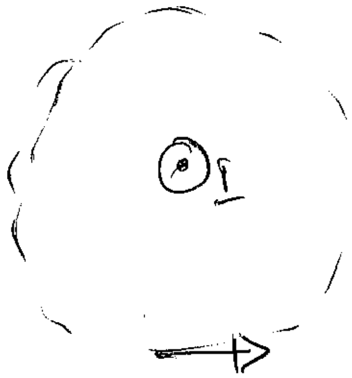
Example 1a

Long thin wire



(We know already! $B = \frac{\mu_0 I}{2\pi d}$)

Top view



← Amperean loop/curve

By symmetry, $\vec{B}$ points along curve and is the same at all points on curve

$$\oint \vec{B} \cdot d\vec{l}$$

$$= \oint B\,dl = B \oint dl = B \cdot 2\pi d$$

Ampere's law gives $\quad B \cdot 2\pi d = \mu_0 I$

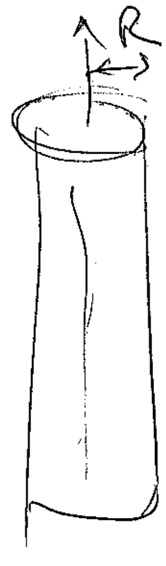
$$\Rightarrow B = \frac{\mu_0 I}{2\pi d}$$

Example 1b

Long thick wire,

cylindrically symmetric current density

J depends only on r

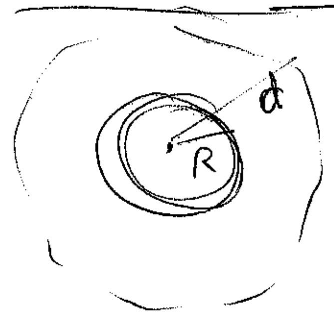


$$I_{encl.} = \int \vec{J}\cdot d\vec{S} = 2\pi \int_0^R r J(r)\, dr$$

$\left\{ dS = 2\pi r\, dr \right)$

~~Eg., if $J(r) = J$ (const), then $I_{encl} = \pi R^2 J$~~

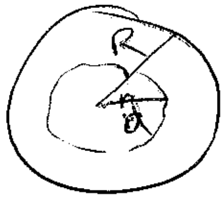
Outside wire $(d > R)$



$$B \cdot 2\pi d = \mu_0 I$$

$$B(d) = \frac{\mu_0 I}{2\pi d}$$ Exactly as for thin wire

Inside wire $(d < R)$



$$B \cdot 2\pi d = \mu_0 I_{enc.}$$

$$= \mu_0 \cdot 2\pi \int_0^d r J(r)\, dr$$

$$B = \frac{\mu_0}{d} \int_0^d r J(r)\, dr$$

If $J(r) =$ constant, then $J = \frac{I}{\pi R^2}$

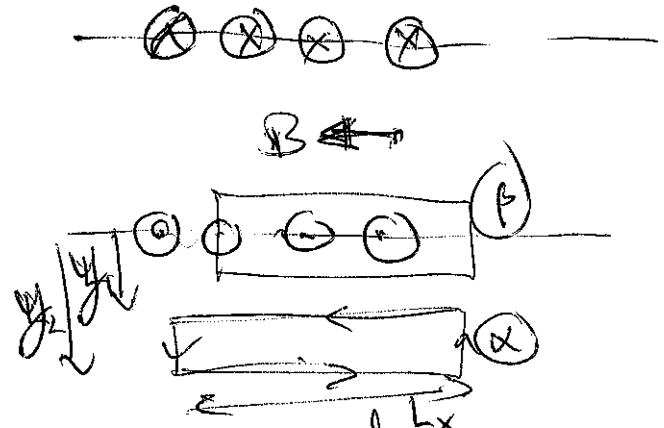
then $$B(d) = \frac{\mu_0}{d} \cdot \frac{I}{\pi R^2} \cdot \frac{d^2}{2} = \frac{\mu_0 I}{2\pi} \frac{d}{R^2}$$

Exercise: Show that inside & outside formulae match at $d = R$.

Infinitely long solenoid

We found, on axis only, using B-S law,

$$B = \mu_0 n I$$



n = turns per unit length

Using symmetry arguments, $\vec{B}$ must point parallel to axis everywhere, leftward inside, rightward outside.

Consider loop outside (α): $\oint \vec{B}\cdot d\vec{l} = B(y_1) L_x - B(y_2) L_x$

$$\Rightarrow \quad B(y_1) = B(y_2) = B(\infty) = 0$$

$\Rightarrow$ Field outside is ZERO !

Consider loop straddling the coil surface (β) :

$$B \cdot L_x = \mu_0 N I$$

$$\Rightarrow \quad B = \mu_0 n I$$ EVERYWHERE INSIDE SOLENOID !

$\Rightarrow$ Solenoid can be used to produce nearly uniform field.

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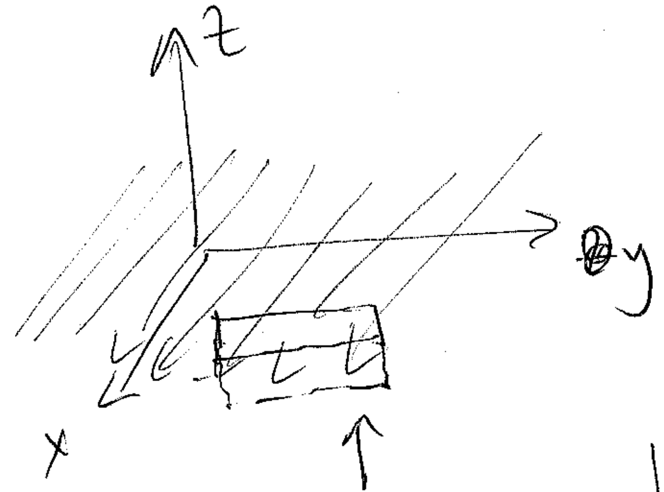
* Infinite surface of current   optional



$$\vec{B} = \frac{\mu_0}{2} K \hat{j} \quad z < 0$$

$$- \frac{\mu_0}{2} K \hat{j} \quad \text{for } z > 0$$

Can prove using such Amperian loop

$$B \cdot L_y + B \cdot L_y = \mu_0 K L_y$$

$$\Rightarrow B = \frac{\mu_0}{2} K$$

Surface current not so strange. Simply think of many parallel wires packed next to each other. If n wires per unit length, $K = nI$